

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury • Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes • It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water • Local people ate large amounts of fish (critical group) so received toxic doses	4
01.2	• Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) • It crosses the placenta and harms the developing unborn baby • Mercury is a teratogen, preventing normal gene expression / development	3
01.3	• Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs • Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut • Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
02.1	• Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned • HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted • A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention)	3

Question	Marking guidance	Marks
02.2	- Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk - Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases	4
02.3	- CFCs are drained from the appliance and incinerated / broken down at high temperature - Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime	2
02.4	CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	
03.1	- Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) - Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river - Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds - Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion - Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind	6
03.2	- Vegetation / dead organic matter under water decays anaerobically - Releasing methane, a powerful greenhouse gas	2
03.3	- After heavy rain surplus water is stored, reducing peak flow / flooding downstream - In dry periods the dam releases water to maintain river levels / supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	
04.1	- Spearman's rank correlation coefficient - Chi-squared test (frequencies / counts in categories) - t-test (two means of measured data) - Spearman's rank correlation coefficient	4
04.2	Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data	1
04.3	- The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level - The negative sign shows a negative correlation: one variable decreases as the other increases - There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation)	3

Question	Marking guidance	Marks
04.4	• Show the standard deviation / range of the original values around each mean • If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
05.1	• Fission: splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons • Fusion: joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars	2
05.2	• Deuterium (hydrogen-2) extracted from (sea)water • Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor)	2
05.3	• Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide • Vacuum so the plasma is not cooled by air • Magnetic field holds the plasma centrally so it does not touch the container walls and cool	3
05.4	• Fuel (deuterium / lithium) is abundant / effectively unlimited • Much less radioactive waste (no long-lived fission products); no chain reaction / meltdown risk; no weapons material • Not yet achieved sustained net energy output: reactors (JET) use more energy than they release; ITER aims for 500 MW out from 50 MW in; laser fusion also experimental; commercial viability decades away	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	
06.1	• Aerobic decomposition produces little methane and returns nutrients to the soil	1
06.2	• It may contain concentrated heavy metals / toxic residues • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble	2
06.3	<i>Any 2 from:</i> • Impermeable polymer / clay liner to stop leachate escaping • Collection and treatment of leachate • Collection of methane and use as fuel • Daily covering of waste / final capping and restoration	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
07.1	• 2020 has higher peak flow in the wet season (32 vs $22 \text{ m}^3 \text{ s}^{-1}$ in March) • 2020 has much lower dry-season flow (2 vs $9 \text{ m}^3 \text{ s}^{-1}$ in July / August) • Flow is much more variable / seasonal in 2020; the river almost dries up	3
07.2	• In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow • With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk • Little stored ice remains to melt in the dry season so flow falls very low	3

Question	Marking guidance	Marks
07.3	<i>Any 2 from:</i> • Water shortages for irrigation / drinking / HEP in the dry season • Flooding in the wet season • Sudden floods from ice-lake outbursts (glacial lake outburst floods)	2
07.4	• Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
08.1	• Solar PV cost fell dramatically from about 480 to 50 \$/MWh (about 90%), the steepest fall • Wind fell steadily from 200 to 45 • Gas stayed roughly constant at 60-70; solar and wind became cheaper than gas around 2015-2020 and are projected to fall further	3
08.2	• Development / research costs are still being paid for and there are no economies of scale in manufacture • They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure • As output grows, mass production, improved designs / efficiency and experience reduce costs	3
08.3	• Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid; running costs are very low • Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments	2
08.4	• Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user • Some impacts (climate change) cannot yet be costed; decommissioning and waste costs are also uncertain	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	
09.1	• $(2900 - 1400) / 1400 \times 100$ • = 107% (accept 107.1)	2
09.2	• Large decrease from 24 per year (1975-79) to 2 per year (2010-14) • Steepest fall between the late 1970s and 1980s; slower decline / levelling off after 1995	2
09.3	<i>Any 3 from:</i> • Double-hulled tankers required for all large tankers • Improved navigation (GPS, AIS) and offshore shipping routes • Inert gas systems reducing explosions • Twin engines / rudders reducing groundings after mechanical failure • Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) • Separate ballast tanks / crude oil washing	3

Question	Marking guidance	Marks
09.4	• Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff • Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) • Growth in oil use and transport means more small releases even if large accidents are rarer	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Uses / benefits: nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • Risks: free radical production, DNA damage, mutation, cancer; acute and chronic, somatic and gonadic effects; contamination vs exposure; activation products; long-lived wastes; accidents (Chernobyl, Fukushima); weapons fallout • Factors affecting risk: type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • Principles: ALARA, BATNEEC; closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • Monitoring: dosimeters, film badges, air and contamination monitors; environmental monitoring via CPA and CGM; waste categories and disposal (vitrification, encapsulation, Drigg) • Evaluation / judgement: benefits generally outweigh risks when controls are applied; public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9
10.2	• Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25